Math 54 — Week 7

Solutions

P1.

Idea. Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 = \mathbf{x}$ by row reducing $[\,\mathbf{b}_1\ \mathbf{b}_2\ \ \mathbf{x}\,]$.

1. Row reduce:
$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 3 & 5 & 9 \end{array}\right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -1 & -3 \end{array}\right]$$
2. Back-substitute: $c_2 = 3$, then $c_1 = 4 - 2c_2 = -2$. So $[\,\mathbf{x}\,]_{\mathcal{B}} = (-2, 3)$.
3. Check:
$$-2\begin{bmatrix}1\\3\end{bmatrix} + 3\begin{bmatrix}2\\5\end{bmatrix} = \begin{bmatrix}-2+6\\-6+15\end{bmatrix} = \begin{bmatrix}4\\9\end{bmatrix}.$$

P2.

Idea. Work with coefficient vectors in the standard basis $\{1, t, t^2\}$; coordinates keep linear relations.

1. (a) Solve $c_1(1+t) + c_2(t+t^2) + c_3(1+t^2) = 3 + 2t - t^2$. Matching the coefficients of 1, t, t^2:
$$c_1 + c_3 = 3, \qquad c_1 + c_2 = 2, \qquad c_2 + c_3 = -1.$$
2. Adding the three equations gives $2(c_1 + c_2 + c_3) = 4$, so $c_1 + c_2 + c_3 = 2$. Subtracting each equation from this: $c_2 = 2 - 3 = -1$, $c_3 = 2 - 2 = 0$, $c_1 = 2 - (-1) = 3$. So $[\,p\,]_{\mathcal{B}} = (3, -1, 0)$. Check: $3(1+t) - (t+t^2) = 3 + 2t - t^2$.
3. (b) The coordinate vectors are $(1, -1, 0)$, $(0, 1, -1)$, $(1, 0, -1)$. The third is the sum of the first two, so the polynomials are dependent:
$$1 - t^2 = (1 - t) + (t - t^2).$$

P3.

Idea. Row reduce $[\,\mathcal{C} \mid \mathcal{B}\,]$ to $[\,I \mid P_{\mathcal{C}\leftarrow\mathcal{B}}\,]$; the other matrix is its inverse.

1. Row reduce:
$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 3 \\ 1 & 1 & 2 & 4 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 1 \end{array}\right], \qquad P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix}1 & 3\\ 1 & 1\end{bmatrix}.$$
2. Its determinant is $1 - 3 = -2$, so by the 2×2 inverse formula
$$P_{\mathcal{B}\leftarrow\mathcal{C}} = \frac{1}{-2}\begin{bmatrix}1 & -3\\ -1 & 1\end{bmatrix} = \begin{bmatrix}-1/2 & 3/2\\ 1/2 & -1/2\end{bmatrix}.$$
3. Then
$$[\,\mathbf{x}\,]_{\mathcal{C}} = \begin{bmatrix}1 & 3\\ 1 & 1\end{bmatrix}\begin{bmatrix}2\\-1\end{bmatrix} = \begin{bmatrix}2-3\\2-1\end{bmatrix} = \begin{bmatrix}-1\\1\end{bmatrix}.$$
4. Check: $\mathbf{x} = 2(1,2) - (3,4) = (-1, 0)$, and $-1(1,1) + 1(0,1) = (-1, 0)$.

P4.

Idea. Multiply $A\mathbf{v}$ and compare with $\mathbf{v}$; for $\lambda = 4$, test whether $A - 4I$ is invertible.

1. By the row-vector rule,
$$A\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}3+0+0\\-1-2+0\\0\end{bmatrix} = \begin{bmatrix}3\\-3\\0\end{bmatrix} = 3\begin{bmatrix}1\\-1\\0\end{bmatrix},$$
so $\mathbf{v}$ is an eigenvector for the eigenvalue 3.

2. $A - 4I$ is below. Expanding down column 2 (one nonzero entry, -2 in position $(2,2)$, sign $+$):
$$A - 4I = \begin{bmatrix}-1 & 0 & 1\\-1 & -2 & -1\\0 & 0 & -3\end{bmatrix}, \qquad \det(A-4I) = -2\det\begin{bmatrix}-1 & 1\\0 & -3\end{bmatrix} = -2\cdot 3 = -6.$$
This is not 0, so $A - 4I$ is invertible and 4 is not an eigenvalue.

P5.

Idea. Column 2 of $A - \lambda I$ has a single nonzero entry; expand down it and keep the factors.

1. Expanding down column 2,
$$\det(A-\lambda I) = (2-\lambda)\det\begin{bmatrix}3-\lambda & 1\\0 & 1-\lambda\end{bmatrix} = (2-\lambda)(3-\lambda)(1-\lambda).$$
The eigenvalues are $1, 2, 3$.

2. For $\lambda = 1$:
$$A - I = \begin{bmatrix}2 & 0 & 1\\-1 & 1 & -1\\0 & 0 & 0\end{bmatrix}.$$
Row 1 gives $x_3 = -2x_1$, and row 2 gives $x_2 = x_1 + x_3 = -x_1$. With x_1 free, a basis of the eigenspace is $\{(1,-1,-2)\}$.

3. Check:
$$A\begin{bmatrix}1\\-1\\-2\end{bmatrix} = \begin{bmatrix}3+0-2\\-1-2+2\\-2\end{bmatrix} = \begin{bmatrix}1\\-1\\-2\end{bmatrix}.$$

P6.

Idea. For a 2×2 matrix the characteristic polynomial is $\lambda^2 - (\operatorname{tr} A)\lambda + \det A$.

1. First matrix: $\operatorname{tr} = 4$ and $\det = 4 - 49 = -45$, so
$$\lambda^2 - 4\lambda - 45 = (\lambda - 9)(\lambda + 5),$$
and the eigenvalues are 9 and -5. Check: $9 + (-5) = 4$ and $9\cdot(-5) = -45$.

2. Second matrix: $\operatorname{tr} = 8$ and $\det = 15 - 12 = 3$, so the characteristic polynomial is $\lambda^2 - 8\lambda + 3$. By the quadratic formula the eigenvalues are
$$\lambda = \frac{8 \pm \sqrt{64-12}}{2} = 4 \pm \sqrt{13}.$$

P7.

Idea. The matrix is upper triangular, so its eigenvalues are the diagonal entries (Theorem 1).

1. The diagonal entries are $2, -1, 2, 0$, so the characteristic polynomial is $(2-\lambda)^2(-1-\lambda)(0-\lambda)$.

2. The eigenvalues are 2 (multiplicity 2), -1 and 0 (multiplicity 1 each).

3. 0 is an eigenvalue, so the matrix is not invertible.

P8.

Idea. Apply A to $A\mathbf{x} = \lambda\mathbf{x}$, or add $3\mathbf{x}$ to it; for the inverse, multiply by A^{-1}.

1. (a) Using $A\mathbf{x} = \lambda\mathbf{x}$ twice, and adding $3\mathbf{x}$:

$$A^2\mathbf{x} = A(\lambda\mathbf{x}) = \lambda A\mathbf{x} = \lambda^2\mathbf{x}, \qquad (A + 3I)\mathbf{x} = \lambda\mathbf{x} + 3\mathbf{x} = (\lambda + 3)\mathbf{x}.$$

   Since $\mathbf{x} \neq \mathbf{0}$, it is an eigenvector of A^2 for λ^2 and of $A + 3I$ for $\lambda + 3$.

2. (b) If $\lambda = 0$, then $A\mathbf{x} = \mathbf{0}$ with $\mathbf{x} \neq \mathbf{0}$, so A would not be invertible. So $\lambda \neq 0$.

3. Multiply $A\mathbf{x} = \lambda\mathbf{x}$ on the left by A^{-1}: $\mathbf{x} = \lambda A^{-1}\mathbf{x}$. Dividing by λ, $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.